

Appendix

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Packages

```
library(tidyverse)
library(VGAM)
library(ordinal)
library(MASS)
library(effects)
library(broom)
library(knitr)
library(kableExtra)
```

Load Data

```
raw <- read.csv("data.csv", na.strings = c("NA", ""), stringsAsFactors = FALSE)
dim(raw)
```

```
## [1] 10259  114
```

Data Cleaning

Select Relevant Variables

```
df <- raw %>%
  dplyr::select(ResponseId, JobSat, AISelect, YearsCodePro, EdLevel,
                Employment, ConvertedCompYearly, RemoteWork, Country)
summary(df)
```

```
##   ResponseId      JobSat      AISelect      YearsCodePro
##   Min.      :    1      Min.      : 0.000      Length:10259      Length:10259
##   1st Qu.: 2566      1st Qu.: 6.000      Class :character      Class :character
##   Median : 5130      Median : 7.000      Mode  :character      Mode  :character
##   Mean    : 5130      Mean    : 6.963
##   3rd Qu.: 7694      3rd Qu.: 8.000
##   Max.    :10259      Max.    :10.000
##                NA's      :4643
##   EdLevel      Employment      ConvertedCompYearly      RemoteWork
##   Length:10259      Length:10259      Min.      :    1      Length:10259
```

```
## Class :character   Class :character   1st Qu.: 37096   Class :character
## Mode :character   Mode :character   Median : 69146   Mode :character
##                                     Mean : 86990
##                                     3rd Qu.: 111417
##                                     Max. :2000000
##                                     NA's :4287
## Country
## Length:10259
## Class :character
## Mode :character
##
##
##
##
```

Response Variable: JobSat -> 3 Ordered Categories

```
table(as.numeric(df$JobSat), useNA = "always")

##
##    0    1    2    3    4    5    6    7    8    9   10 <NA>
##   61   64  139  224  221  390  675 1167 1497  719  459 4643

df <- df %>%
  mutate(
    JobSat_num = as.numeric(JobSat),
    JobSat_cat = case_when(
      JobSat_num <= 5 ~ "Low",
      JobSat_num <= 7 ~ "Moderate",
      JobSat_num <= 10 ~ "High",
      TRUE ~ NA_character_
    ),
    JobSat_cat = fct_relevel(JobSat_cat, "Low", "Moderate") # High is last/highest
  )

table(df$JobSat_cat, useNA = "always")

##
##      Low Moderate      High      <NA>
##    1099      1842      2675      4643
```

YearsCodePro: Recode Text Entries and Center

```
unique(df$YearsCodePro[grepl("year", df$YearsCodePro, ignore.case = TRUE)])

## [1] "Less than 1 year" "More than 50 years"

df <- df %>%
  mutate(
    YearsCodePro = case_when(
      YearsCodePro == "Less than 1 year" ~ "0.5",
      YearsCodePro == "More than 50 years" ~ "50",
      TRUE ~ YearsCodePro
    ),
    YearsCodePro = as.numeric(YearsCodePro)
```

```
)

summary(df$YearsCodePro)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
##      0.50    4.00    8.00   10.84   15.00   50.00   1196

mean(df$YearsCodePro, na.rm = TRUE)

## [1] 10.83946

# Center around mean (19.2 years)
df <- df %>% mutate(CYears = YearsCodePro - mean(YearsCodePro, na.rm = TRUE))
```

Annual Compensation: Log-Transform and Center

Salary is right-skewed with values up to \$2M. Log-transforming then center.

```
sum(df$ConvertedCompYearly > 500000, na.rm = TRUE) # 25 above $500k

## [1] 25

sum(df$ConvertedCompYearly > 1000000, na.rm = TRUE) # 9 above $1M

## [1] 9

df <- df %>% mutate(log_comp = log(ConvertedCompYearly))

summary(df$log_comp)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
##      0.00   10.52   11.14   10.87   11.62   14.51   4287

mean(df$log_comp, na.rm = TRUE)

## [1] 10.86988

# Center around mean log-comp
df <- df %>% mutate(Clog_comp = log_comp - mean(log_comp, na.rm = TRUE))
```

Education Level: Collapsing and Level

```
df <- df %>%
  mutate(
    EdLevel = case_when(
      grepl("Bachelor", EdLevel) ~ "Bachelor's",
      grepl("Master", EdLevel) ~ "Master's",
      grepl("Professional degree|Ph\\.D", EdLevel, ignore.case = TRUE) ~ "Professional/PhD",
      grepl("Some college|Associate|Secondary|Primary|Something",
        EdLevel, ignore.case = TRUE) ~ "No Degree",
      TRUE ~ NA_character_
    ),
    EdLevel = fct_relevel(EdLevel, "No Degree", "Bachelor's", "Master's")
  )

table(df$EdLevel, useNA = "always")

##
```

```
##          No Degree          Bachelor's          Master's Professional/PhD
##          2839             4253             2662             505
##          <NA>
##          0
```

Employment Type: Simplifying

```
df <- df %>%
  mutate(
    Employment = case_when(
      grepl("Employed, full-time", Employment) ~ "Full-time",
      grepl("Independent contractor|freelancer|self-employed",
            Employment, ignore.case = TRUE) ~ "Freelance",
      grepl("Student", Employment, ignore.case = TRUE) ~ "Student",
      grepl("part-time", Employment, ignore.case = TRUE) ~ "Part-time",
      grepl("Not employed", Employment, ignore.case = TRUE) ~ "Not employed",
      TRUE ~ "Other"
    ),
    Employment = fct_relevel(Employment, "Full-time")
  )

table(df$Employment, useNA = "always")
```

```
##
## Full-time  Freelance Not employed  Other  Part-time  Student
##      7343      1106      462      173      195      980
##      <NA>
##      0
```

AI Tool Adoption

```
df <- df %>%
  mutate(
    AISelect = case_when(
      AISelect == "Yes" ~ "Uses AI",
      AISelect == "No, but I plan to soon" ~ "Plans to use",
      AISelect == "No, and I don't plan to" ~ "Does not use",
      TRUE ~ NA_character_
    ),
    AISelect = fct_relevel(AISelect, "Does not use", "Plans to use")
  )

table(df$AISelect, useNA = "always")
```

```
##
## Does not use Plans to use  Uses AI  <NA>
##      2797      1384      6077      1
```

Remote Work

```
df <- df %>%
  mutate(
    RemoteWork = case_when(
```

```

RemoteWork == "In-person" ~ "In-person",
RemoteWork == "Hybrid (some remote, some in-person)" ~ "Hybrid",
RemoteWork == "Remote" ~ "Remote",
TRUE ~ NA_character_
),
RemoteWork = fct_relevel(RemoteWork, "In-person", "Hybrid")
)

table(df$RemoteWork, useNA = "always")

```

```

##
## In-person   Hybrid   Remote   <NA>
##      1641     3552     3674     1392

```

Remove Invalid Country Entries

```
df <- df %>% filter(!Country %in% c("Nomadic", ""))
```

Complete Case Analysis

```

df_model <- df %>%
  dplyr::select(ResponseId, JobSat_cat, AISelect, CYears, EdLevel,
                Employment, Clog_comp, RemoteWork, Country) %>%
  drop_na()

cat("Original rows:      ", nrow(df), "\n")

## Original rows:      10253

cat("Complete case rows:", nrow(df_model), "\n")

## Complete case rows: 4129

cat("Rows dropped:      ", nrow(df) - nrow(df_model),
    sprintf("(%.1f%%)\n", (nrow(df) - nrow(df_model)) / nrow(df) * 100))

## Rows dropped:      6124 (59.7%)

```

Final Dataset Summary

```
summary(df_model)
```

```

##      ResponseId      JobSat_cat      AISelect      CYears
## Min.   : 73      Low      : 804      Does not use:1008      Min.   : -10.339
## 1st Qu.: 3050      Moderate:1334      Plans to use: 555      1st Qu.: -6.839
## Median : 5468      High      :1991      Uses AI      :2566      Median : -1.839
## Mean   : 5424                                     Mean   : -0.151
## 3rd Qu.: 7848                                     3rd Qu.:  4.161
## Max.   :10257                                     Max.   : 39.161
##      EdLevel      Employment      Clog_comp
## No Degree      : 908      Full-time :3739      Min.   : -10.86988
## Bachelor's     :1887      Freelance  : 235      1st Qu.: -0.32049
## Master's       :1160      Not employed:  0      Median :  0.28371
## Professional/PhD: 174      Other      :  0      Mean   :  0.03484

```

```
##          Part-time : 80  3rd Qu.: 0.75116
##          Student   : 75  Max.    : 3.49261
## RemoteWork Country
## In-person: 657 Length:4129
## Hybrid   :1698 Class :character
## Remote   :1774 Mode  :character
##
##
##
```

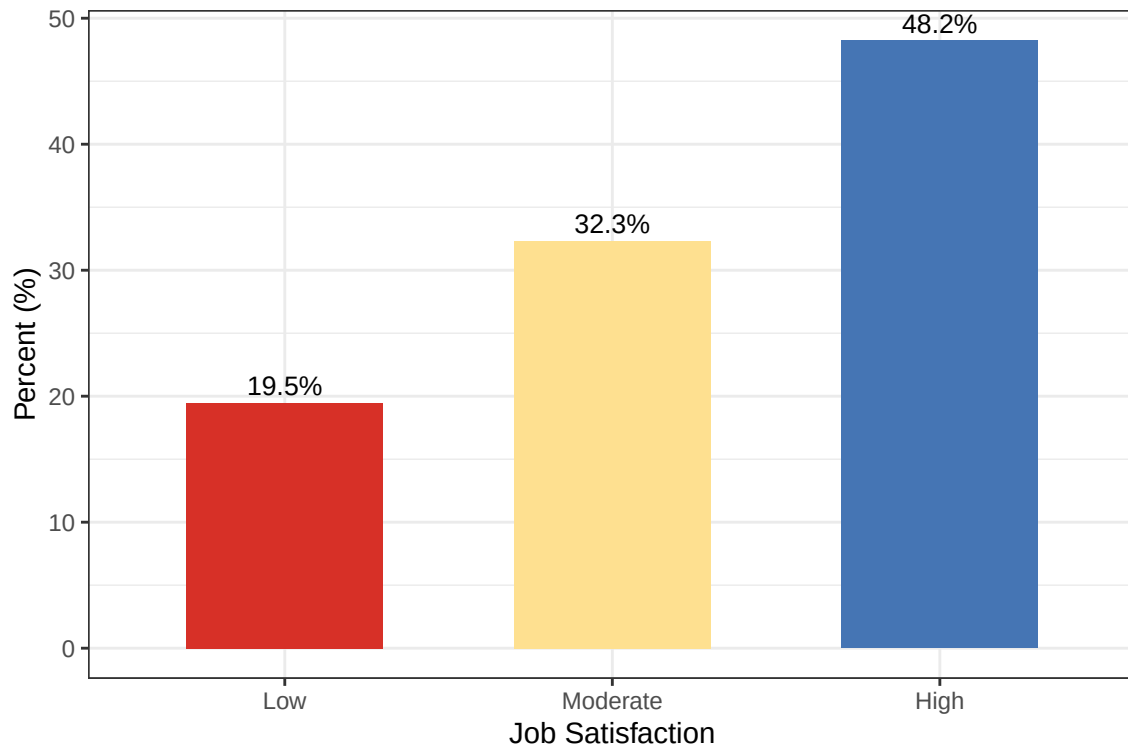
Exploratory Data Analysis

Response: Distribution of Job Satisfaction

```
table(df_model$JobSat_cat)
```

```
##
##      Low Moderate      High
##      804      1334      1991
```

```
df_model %>%
  count(JobSat_cat) %>%
  mutate(pct = n / sum(n) * 100) %>%
  ggplot(aes(x = JobSat_cat, y = pct, fill = JobSat_cat)) +
  geom_col(width = 0.6) +
  geom_text(aes(label = sprintf("%.1f%%", pct)), vjust = -0.4, size = 3.5) +
  scale_fill_manual(values = c("Low" = "#d73027", "Moderate" = "#fee090", "High" = "#4575b4")) +
  labs(x = "Job Satisfaction", y = "Percent (%)") +
  theme_bw() + theme(legend.position = "none")
```



Cumulative Log Odds EDA: Categorical Predictors

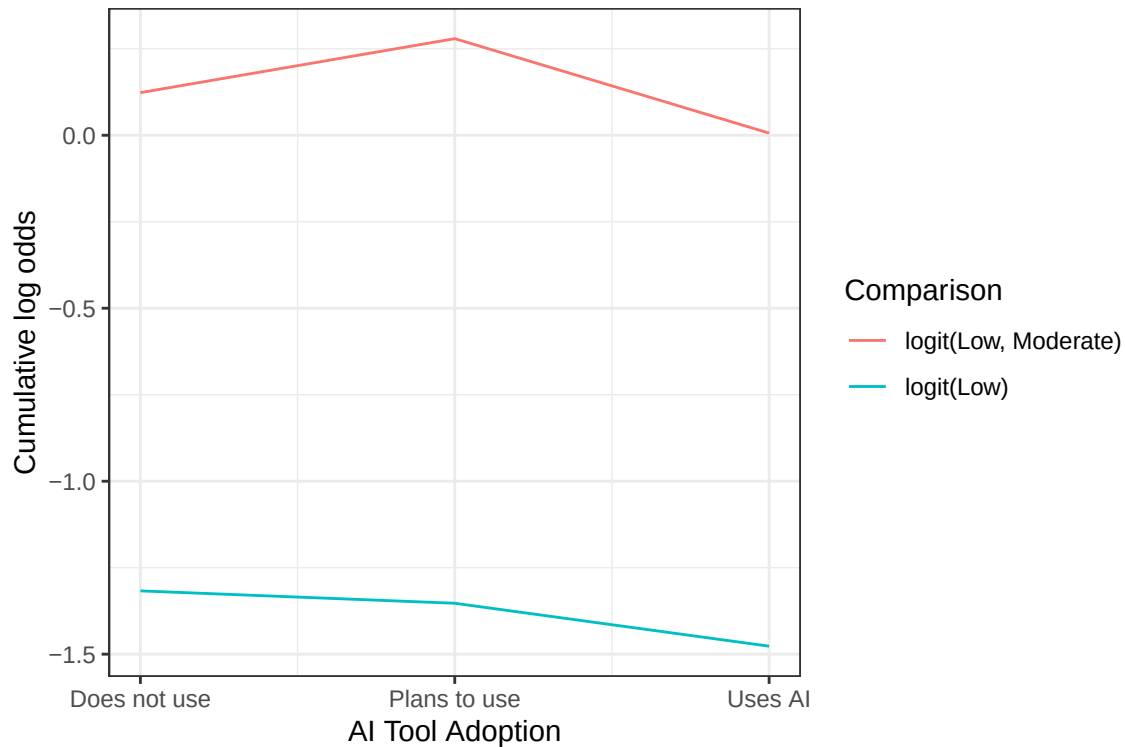
AI Tool Adoption

```
ai_lodds <- df_model %>%
  group_by(AISelect) %>%
  summarize(
    prob1 = mean(JobSat_cat == "Low"),
    lodds1 = log(prob1 / (1 - prob1)),
    prob12 = mean(JobSat_cat %in% c("Low", "Moderate")),
    lodds12 = log(prob12 / (1 - prob12))
  )
```

```
ai_lodds
```

```
## # A tibble: 3 x 5
##   AISelect      prob1 lodds1 prob12 lodds12
##   <fct>      <dbl>  <dbl>  <dbl>  <dbl>
## 1 Does not use 0.211  -1.32  0.531  0.123
## 2 Plans to use 0.205  -1.35  0.569  0.279
## 3 Uses AI      0.186  -1.48  0.502  0.00624
```

```
ggplot(ai_lodds, aes(x = as.numeric(AISelect))) +
  geom_line(aes(y = lodds1, color = "logit(Low)")) +
  geom_line(aes(y = lodds12, color = "logit(Low, Moderate)")) +
  scale_x_continuous(breaks = 1:3, labels = levels(df_model$AISelect)) +
  labs(x = "AI Tool Adoption", y = "Cumulative log odds", color = "Comparison") +
  theme_bw()
```



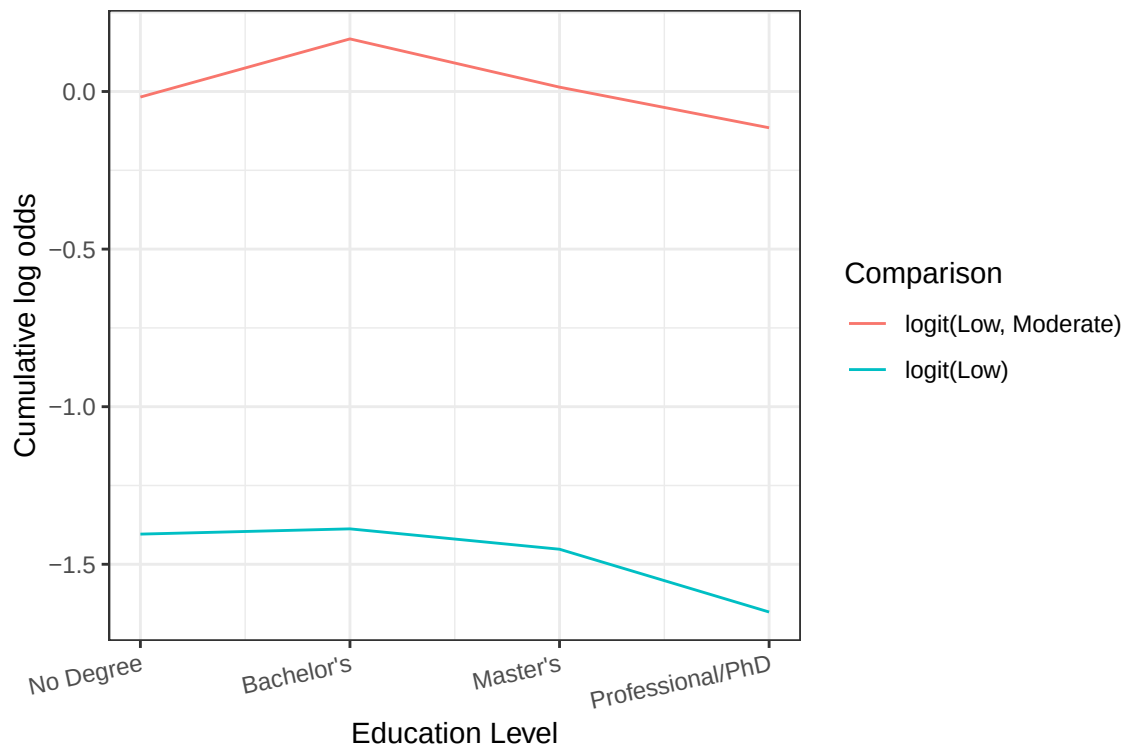
Education Level

```
ed_lodds <- df_model %>%
  group_by(EdLevel) %>%
  summarize(
    prob1 = mean(JobSat_cat == "Low"),
    lodds1 = log(prob1 / (1 - prob1)),
    prob12 = mean(JobSat_cat %in% c("Low", "Moderate")),
    lodds12 = log(prob12 / (1 - prob12))
  )

ed_lodds

## # A tibble: 4 x 5
##   EdLevel      prob1 lodds1 prob12 lodds12
##   <fct>      <dbl> <dbl> <dbl> <dbl>
## 1 No Degree    0.197  -1.40  0.496 -0.0176
## 2 Bachelor's  0.200  -1.39  0.542  0.167
## 3 Master's    0.190  -1.45  0.503  0.0138
## 4 Professional/PhD 0.161  -1.65  0.471 -0.115

ggplot(ed_lodds, aes(x = as.numeric(EdLevel))) +
  geom_line(aes(y = lodds1, color = "logit(Low)")) +
  geom_line(aes(y = lodds12, color = "logit(Low, Moderate)")) +
  scale_x_continuous(breaks = 1:4, labels = levels(df_model$EdLevel)) +
  labs(x = "Education Level", y = "Cumulative log odds", color = "Comparison") +
  theme_bw() +
  theme(axis.text.x = element_text(angle = 12, hjust = 1))
```

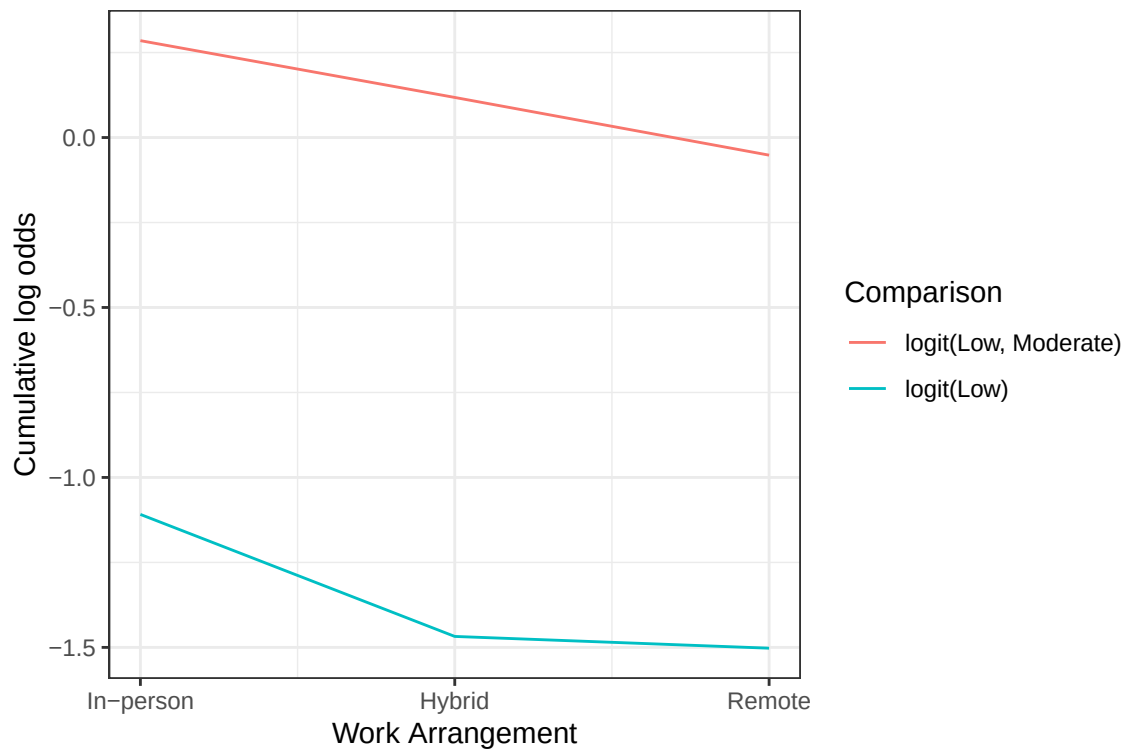
Remote Work Status

```
remote_lodds <- df_model %>%
  group_by(RemoteWork) %>%
  summarize(
    prob1 = mean(JobSat_cat == "Low"),
    lodds1 = log(prob1 / (1 - prob1)),
    prob12 = mean(JobSat_cat %in% c("Low", "Moderate")),
    lodds12 = log(prob12 / (1 - prob12))
  )
```

```
remote_lodds
```

```
## # A tibble: 3 x 5
##   RemoteWork prob1 lodds1 prob12 lodds12
##   <fct>      <dbl> <dbl> <dbl> <dbl>
## 1 In-person  0.248  -1.11  0.571  0.285
## 2 Hybrid    0.187  -1.47  0.529  0.118
## 3 Remote    0.182  -1.50  0.487 -0.0519
```

```
ggplot(remote_lodds, aes(x = as.numeric(RemoteWork))) +
  geom_line(aes(y = lodds1, color = "logit(Low)")) +
  geom_line(aes(y = lodds12, color = "logit(Low, Moderate)")) +
  scale_x_continuous(breaks = 1:3, labels = levels(df_model$RemoteWork)) +
  labs(x = "Work Arrangement", y = "Cumulative log odds", color = "Comparison") +
  theme_bw()
```



Cumulative Log Odds EDA: Quantitative Predictors

Group respondents into 5 quantiles, compute mean predictor value and cumulative log odds per group. Check for linearity and parallel trends.

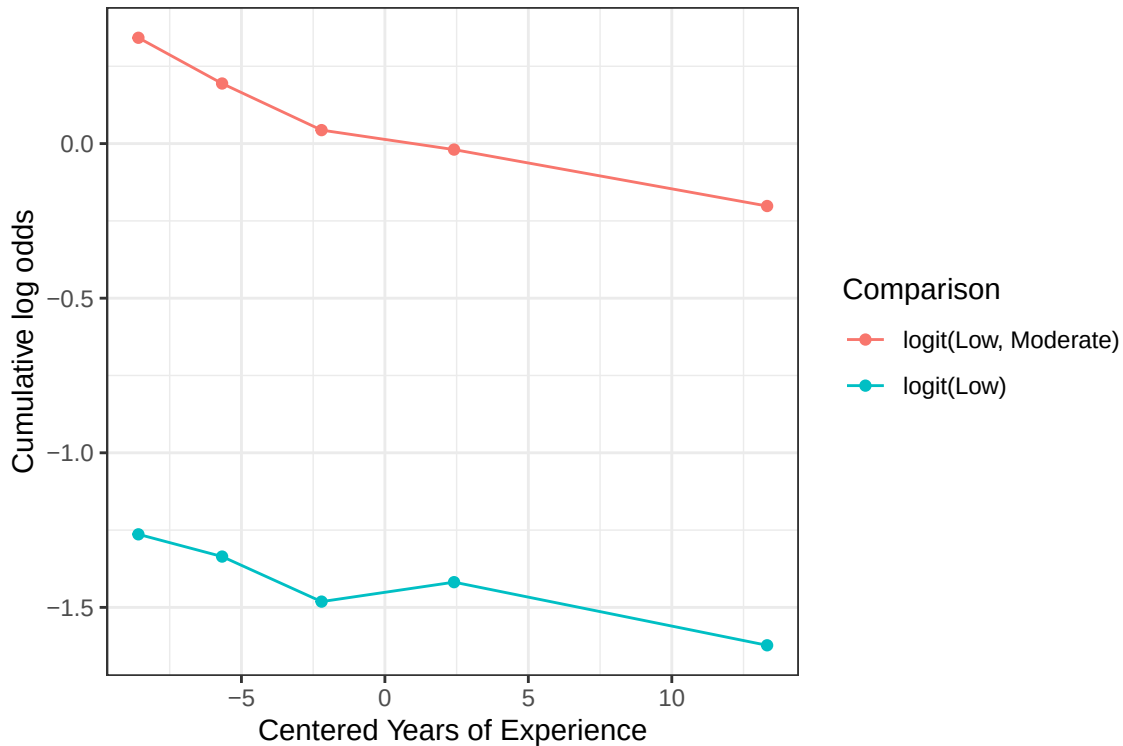
Years of Professional Experience

```
exp_lodds <- df_model %>%
  mutate(exp_grp = ntile(CYears, 5)) %>%
  group_by(exp_grp) %>%
  summarize(
    mean_exp = mean(CYears),
    prob1 = mean(JobSat_cat == "Low"),
    lodds1 = log(prob1 / (1 - prob1)),
    prob12 = mean(JobSat_cat %in% c("Low", "Moderate")),
    lodds12 = log(prob12 / (1 - prob12))
  )
```

exp_lodds

```
## # A tibble: 5 x 6
##   exp_grp mean_exp prob1 lodds1 prob12 lodds12
##   <int>     <dbl> <dbl>  <dbl>  <dbl>    <dbl>
## 1      1     -8.59  0.220  -1.26  0.585    0.342
## 2      2     -5.67  0.208  -1.34  0.548    0.194
## 3      3     -2.21  0.185  -1.48  0.511    0.0436
## 4      4      2.41  0.195  -1.42  0.495   -0.0194
## 5      5     13.3  0.165  -1.62  0.450   -0.202
```

```
ggplot(exp_lodds, aes(x = mean_exp)) +
  geom_line(aes(y = lodds1, color = "logit(Low)")) +
  geom_line(aes(y = lodds12, color = "logit(Low, Moderate)")) +
  geom_point(aes(y = lodds1, color = "logit(Low)")) +
  geom_point(aes(y = lodds12, color = "logit(Low, Moderate)")) +
  labs(x = "Centered Years of Experience", y = "Cumulative log odds", color = "Comparison") +
  theme_bw()
```



Log Annual Compensation

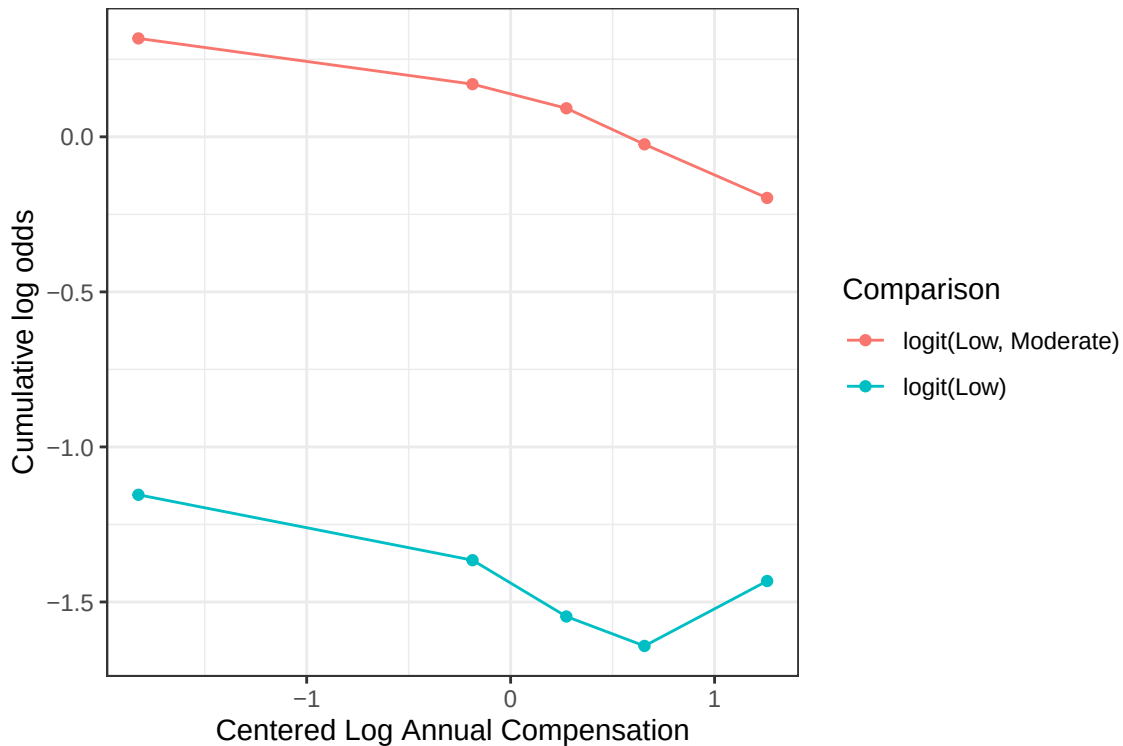
```
comp_lodds <- df_model %>%
  mutate(comp_grp = ntile(Clog_comp, 5)) %>%
  group_by(comp_grp) %>%
  summarize(
    mean_comp = mean(Clog_comp),
    prob1 = mean(JobSat_cat == "Low"),
    lodds1 = log(prob1 / (1 - prob1)),
    prob12 = mean(JobSat_cat %in% c("Low", "Moderate")),
    lodds12 = log(prob12 / (1 - prob12))
  )
```

comp_lodds

```
## # A tibble: 5 x 6
##   comp_grp mean_comp prob1 lodds1 prob12 lodds12
##   <int>     <dbl> <dbl> <dbl> <dbl>    <dbl>
## 1       1    -1.83  0.240 -1.15  0.579  0.317
## 2       2    -0.187 0.203 -1.37  0.542  0.170
## 3       3     0.273 0.176 -1.55  0.523  0.0921
```

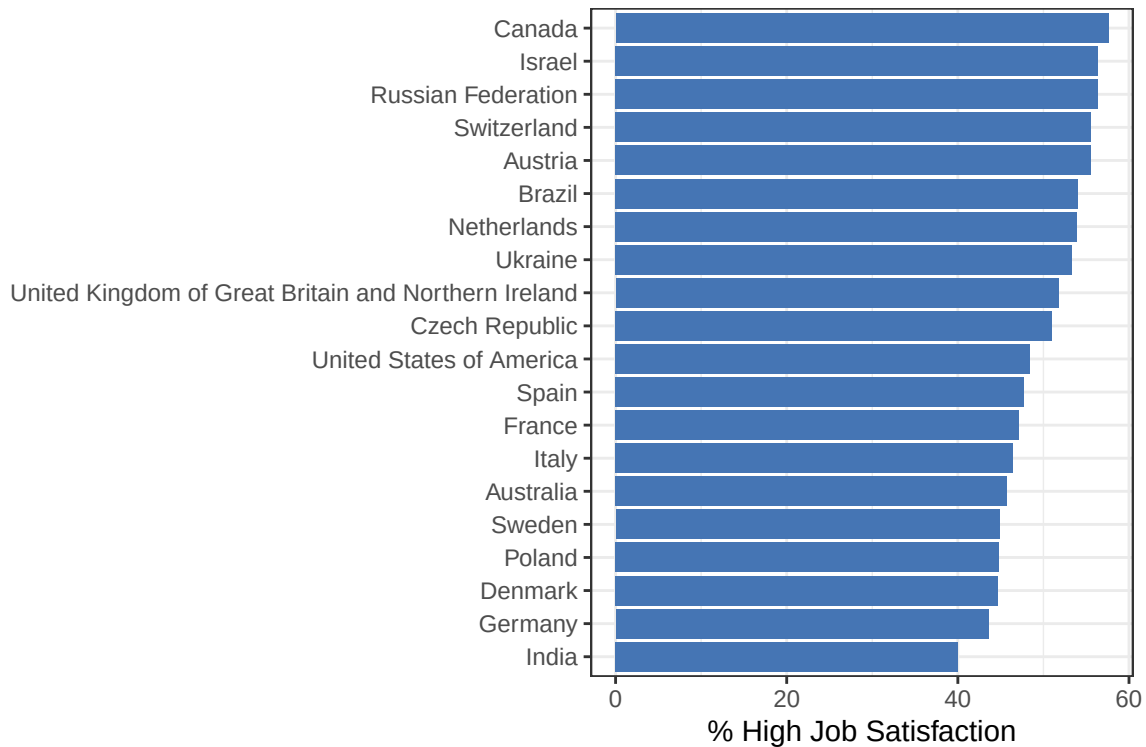
```
## 4      4      0.657 0.162 -1.64  0.494 -0.0242
## 5      5      1.26  0.193 -1.43  0.451 -0.197
```

```
ggplot(comp_lodds, aes(x = mean_comp)) +
  geom_line(aes(y = lodds1, color = "logit(Low)")) +
  geom_line(aes(y = lodds12, color = "logit(Low, Moderate)")) +
  geom_point(aes(y = lodds1, color = "logit(Low)")) +
  geom_point(aes(y = lodds12, color = "logit(Low, Moderate)")) +
  labs(x = "Centered Log Annual Compensation", y = "Cumulative log odds",
       color = "Comparison") +
  theme_bw()
```



Country-Level Variation

```
df_model %>%
  group_by(Country) %>%
  summarise(n = n(), pct_high = mean(JobSat_cat == "High") * 100) %>%
  filter(n >= 20) %>%
  slice_max(n, n = 20) %>%
  ggplot(aes(x = reorder(Country, pct_high), y = pct_high)) +
  geom_col(fill = "#4575b4") +
  coord_flip() +
  labs(x = NULL, y = "% High Job Satisfaction") +
  theme_bw()
```



Modeling

Null CLMM: Estimating ICC

```
m0 <- clmm(JobSat_cat ~ 1 + (1 | Country), data = df_model, Hess = TRUE)
summary(m0)
```

```
## Cumulative Link Mixed Model fitted with the Laplace approximation
##
## formula: JobSat_cat ~ 1 + (1 | Country)
## data:    df_model
##
## link threshold nobs logLik AIC niter max.grad cond.H
## logit flexible 4129 -4274.46 8554.93 119(258) 2.52e-04 6.1e+02
##
## Random effects:
## Groups Name Variance Std.Dev.
## Country (Intercept) 0.008996 0.09485
## Number of groups: Country 125
##
## No Coefficients
##
## Threshold coefficients:
## Estimate Std. Error z value
## Low|Moderate -1.42362 0.04414 -32.25
## Moderate|High 0.06984 0.03715 1.88
```

```
var_u <- as.numeric(VarCorr(m0)$Country)
icc <- var_u / (var_u + pi^2 / 3)
cat(sprintf("ICC: %.3f - %.1f%% of variation is between countries\n", icc, icc * 100))
```

```
## ICC: 0.003 - 0.3% of variation is between countries
```

Main Effects Model (Model 1)

```
m1 <- clmm(JobSat_cat ~ ASelect + CYears + EdLevel +
            Employment + Clog_comp + RemoteWork + (1 | Country),
            data = df_model, Hess = TRUE)
summary(m1)
```

```
## Cumulative Link Mixed Model fitted with the Laplace approximation
##
## formula: JobSat_cat ~ ASelect + CYears + EdLevel + Employment + Clog_comp +
##          RemoteWork + (1 | Country)
## data:    df_model
##
## link threshold nobs logLik AIC niter max.grad cond.H
## logit flexible 4129 -4235.03 8500.05 1705(3605) 1.57e-02 1.4e+04
##
## Random effects:
## Groups Name Variance Std.Dev.
## Country (Intercept) 0.01078 0.1038
## Number of groups: Country 125
##
## Coefficients:
## Estimate Std. Error z value Pr(>|z|)
## ASelectPlans to use -0.046310 0.099544 -0.465 0.641774
## ASelectUses AI 0.204743 0.071895 2.848 0.004402 **
## CYears 0.018328 0.003993 4.590 4.43e-06 ***
## EdLevelBachelor's -0.112031 0.077841 -1.439 0.150085
## EdLevelMaster's -0.012991 0.085858 -0.151 0.879736
## EdLevelProfessional/PhD 0.127794 0.160330 0.797 0.425412
## EmploymentFreelance -0.085998 0.132630 -0.648 0.516720
## EmploymentPart-time 0.346792 0.223849 1.549 0.121328
## EmploymentStudent 0.219553 0.229206 0.958 0.338121
## Clog_comp 0.093759 0.027960 3.353 0.000799 ***
## RemoteWorkHybrid 0.136291 0.087955 1.550 0.121248
## RemoteWorkRemote 0.245632 0.089428 2.747 0.006020 **
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Threshold coefficients:
## Estimate Std. Error z value
## Low|Moderate -1.2339 0.1122 -10.997
## Moderate|High 0.2820 0.1101 2.562
```

Test Random Effect

Comparing Model 1 (with country RE) to fixed-effects-only model via LRT. Boundary LRT

```
m1_fixed <- clm(JobSat_cat ~ ASelect + CYears + EdLevel +
  Employment + Clog_comp + RemoteWork,
  data = df_model)
```

```
lrt_obs <- as.numeric(2 * (logLik(m1) - logLik(m1_fixed)))
lrt_obs
```

```
## [1] 3.060083
```

```
# boundary-corrected p-value
```

```
.5*(1 - pchisq(lrt_obs, 1)) + .5*(1 - pchisq(lrt_obs, 0))
```

```
## [1] 0.04011878
```

Proportional Odds Check

Testing whether the proportional odds assumption holds for key predictors by fitting a partial proportional odds model (parallel = FALSE for ASelect).

```
# Fixed effects partial PO model via VGAM for the PO test
```

```
m1_vglm <- vglm(JobSat_cat ~ ASelect + CYears + EdLevel +
  Employment + Clog_comp + RemoteWork,
  data = df_model, family = cumulative(parallel = TRUE))
```

```
m1_partialPO <- vglm(JobSat_cat ~ ASelect + CYears + EdLevel +
  Employment + Clog_comp + RemoteWork,
  data = df_model,
  family = cumulative(parallel = FALSE ~ ASelect))
```

```
lrtest(m1_partialPO, m1_vglm) # H0: PO holds for ASelect
```

```
## Likelihood ratio test
```

```
##
```

```
## Model 1: JobSat_cat ~ ASelect + CYears + EdLevel + Employment + Clog_comp +
## RemoteWork
```

```
## Model 2: JobSat_cat ~ ASelect + CYears + EdLevel + Employment + Clog_comp +
## RemoteWork
```

```
## #Df LogLik Df Chisq Pr(>Chisq)
```

```
## 1 8242 -4235.3
```

```
## 2 8244 -4236.6 2 2.4499 0.2938
```

```
data.frame(model = c("m1_vglm", "m1_partialPO"),
  AIC = c(AIC(m1_vglm), AIC(m1_partialPO)))
```

```
## model AIC
```

```
## 1 m1_vglm 8501.113
```

```
## 2 m1_partialPO 8502.663
```

Interaction Models

Model 2: AI × Experience

```
m2 <- clmm(JobSat_cat ~ ASelect * CYears + EdLevel +
  Employment + Clog_comp + RemoteWork + (1 | Country),
  data = df_model, Hess = TRUE)
summary(m2)
```

```
## Cumulative Link Mixed Model fitted with the Laplace approximation
##
## formula: JobSat_cat ~ AISelect * CYears + EdLevel + Employment + Clog_comp +
##          RemoteWork + (1 | Country)
## data:    df_model
##
## link threshold nobs logLik   AIC      niter      max.grad cond.H
## logit flexible 4129 -4233.91 8501.82 2223(4604) 1.89e-02 2.0e+04
##
## Random effects:
## Groups Name          Variance Std.Dev.
## Country (Intercept) 0.01019  0.1009
## Number of groups: Country 125
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## AISelectPlans to use   -0.048995   0.099994  -0.490  0.62415
## AISelectUses AI         0.204736   0.072129   2.838  0.00453 **
## CYears                 0.014319   0.006573   2.178  0.02938 *
## EdLevelBachelor's      -0.113942   0.077899  -1.463  0.14355
## EdLevelMaster's        -0.014222   0.085863  -0.166  0.86844
## EdLevelProfessional/PhD 0.121024   0.160411   0.754  0.45057
## EmploymentFreelance    -0.084227   0.132707  -0.635  0.52563
## EmploymentPart-time     0.347382   0.223847   1.552  0.12069
## EmploymentStudent       0.227525   0.229087   0.993  0.32062
## Clog_comp               0.089615   0.028112   3.188  0.00143 **
## RemoteWorkHybrid        0.138154   0.087933   1.571  0.11615
## RemoteWorkRemote        0.243849   0.089423   2.727  0.00639 **
## AISelectPlans to use:CYears -0.005609   0.012161  -0.461  0.64463
## AISelectUses AI:CYears    0.009083   0.008311   1.093  0.27445
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Threshold coefficients:
##              Estimate Std. Error z value
## Low|Moderate  -1.2407    0.1124 -11.038
## Moderate|High  0.2755    0.1103   2.499
```

```
anova(m1, m2) # LRT: does interaction improve fit?
```

```
## Likelihood ratio tests of cumulative link models:
##
## formula:
## m1 JobSat_cat ~ AISelect + CYears + EdLevel + Employment + Clog_comp + RemoteWork + (1 | Country)
## m2 JobSat_cat ~ AISelect * CYears + EdLevel + Employment + Clog_comp + RemoteWork + (1 | Country)
## link: threshold:
## m1 logit flexible
## m2 logit flexible
##
## no.par   AIC logLik LR.stat df Pr(>Chisq)
## m1      15 8500.1 -4235.0
## m2      17 8501.8 -4233.9  2.2309 2    0.3278
```


Model 3: AI × Education

```
m3 <- clmm(JobSat_cat ~ ASelect * EdLevel + CYears +
            Employment + Clog_comp + RemoteWork + (1 | Country),
            data = df_model, Hess = TRUE)
summary(m3)
```

```
## Cumulative Link Mixed Model fitted with the Laplace approximation
##
## formula: JobSat_cat ~ ASelect * EdLevel + CYears + Employment + Clog_comp +
##          RemoteWork + (1 | Country)
## data:    df_model
##
## link threshold nobs logLik   AIC      niter      max.grad cond.H
## logit flexible  4129 -4232.56 8507.11 2828(5869) 6.02e-04 3.4e+04
##
## Random effects:
## Groups Name          Variance Std.Dev.
## Country (Intercept) 0.01116  0.1056
## Number of groups:  Country 125
##
## Coefficients:
##                                     Estimate Std. Error z value
## ASelectPlans to use                -0.127002   0.224484  -0.566
## ASelectUses AI                     0.160700   0.153280   1.048
## EdLevelBachelor's                 -0.215713   0.157580  -1.369
## EdLevelMaster's                   -0.008407   0.174688  -0.048
## EdLevelProfessional/PhD           0.319650   0.300235   1.065
## CYears                           0.018289   0.003998   4.575
## EmploymentFreelance               -0.086009   0.132701  -0.648
## EmploymentPart-time               0.341551   0.224321   1.523
## EmploymentStudent                 0.210728   0.229383   0.919
## Clog_comp                        0.092587   0.028056   3.300
## RemoteWorkHybrid                  0.136518   0.088120   1.549
## RemoteWorkRemote                  0.248275   0.089575   2.772
## ASelectPlans to use:EdLevelBachelor's 0.208377   0.268544   0.776
## ASelectUses AI:EdLevelBachelor's    0.121899   0.184395   0.661
## ASelectPlans to use:EdLevelMaster's  0.053438   0.287359   0.186
## ASelectUses AI:EdLevelMaster's     -0.018842   0.204571  -0.092
## ASelectPlans to use:EdLevelProfessional/PhD -0.875637   0.550732  -1.590
## ASelectUses AI:EdLevelProfessional/PhD -0.157529   0.364890  -0.432
##                                     Pr(>|z|)
## ASelectPlans to use                0.571563
## ASelectUses AI                     0.294453
## EdLevelBachelor's                 0.171026
## EdLevelMaster's                   0.961615
## EdLevelProfessional/PhD           0.287027
## CYears                           4.76e-06 ***
## EmploymentFreelance               0.516893
## EmploymentPart-time               0.127860
## EmploymentStudent                 0.358267
## Clog_comp                        0.000967 ***
## RemoteWorkHybrid                  0.121326
## RemoteWorkRemote                  0.005576 **
```

```
## AISelectPlans to use:EdLevelBachelor's      0.437777
## AISelectUses AI:EdLevelBachelor's          0.508563
## AISelectPlans to use:EdLevelMaster's        0.852475
## AISelectUses AI:EdLevelMaster's            0.926614
## AISelectPlans to use:EdLevelProfessional/PhD 0.111846
## AISelectUses AI:EdLevelProfessional/PhD     0.665948
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Threshold coefficients:
##           Estimate Std. Error z value
## Low|Moderate  -1.2719    0.1519  -8.372
## Moderate|High   0.2456    0.1502   1.634
anova(m1, m3)

## Likelihood ratio tests of cumulative link models:
##
## formula:
## m1 JobSat_cat ~ AISelect + CYears + EdLevel + Employment + Clog_comp + RemoteWork + (1 | Country)
## m3 JobSat_cat ~ AISelect * EdLevel + CYears + Employment + Clog_comp + RemoteWork + (1 | Country)
## link: threshold:
## m1 logit flexible
## m3 logit flexible
##
## no.par    AIC  logLik LR.stat df Pr(>Chisq)
## m1       15 8500.1 -4235.0
## m3       21 8507.1 -4232.6  4.9419 6    0.5513
```

Model Comparison

```
AIC(m0, m1, m2, m3)
```

```
##      df      AIC
## m0   3 8554.925
## m1  15 8500.053
## m2  17 8501.822
## m3  21 8507.111
```

Final Model

```
m_final <- m1  # update to m2 or m3 if interaction is significant
coef(summary(m_final))
```

```
##           Estimate Std. Error z value Pr(>|z|)
## Low|Moderate    -1.23394721 0.112206671 -10.9970931 3.946492e-28
## Moderate|High     0.28199714 0.110083749   2.5616600 1.041732e-02
## AISelectPlans to use -0.04630970 0.099543665  -0.4652199 6.417740e-01
## AISelectUses AI     0.20474326 0.071895240   2.8477999 4.402259e-03
## CYears            0.01832828 0.003993069   4.5900240 4.431950e-06
## EdLevelBachelor's  -0.11203061 0.077840504  -1.4392329 1.500846e-01
## EdLevelMaster's    -0.01299065 0.085858232  -0.1513035 8.797363e-01
## EdLevelProfessional/PhD 0.12779375 0.160329975   0.7970671 4.254120e-01
## EmploymentFreelance -0.08599832 0.132629751  -0.6484090 5.167205e-01
```

```
## EmploymentPart-time      0.34679249 0.223849339 1.5492228 1.213282e-01
## EmploymentStudent        0.21955273 0.229206034 0.9578837 3.381214e-01
## Clog_comp                 0.09375947 0.027960374 3.3532981 7.985464e-04
## RemoteWorkHybrid         0.13629073 0.087954713 1.5495558 1.212482e-01
## RemoteWorkRemote         0.24563151 0.089428294 2.7466867 6.020060e-03
```

```
confint(m_final)
```

```
##              2.5 %      97.5 %
## Low|Moderate    -1.45386824 -1.01402617
## Moderate|High   0.06623695  0.49775732
## AISelectPlans to use -0.24141169 0.14879230
## AISelectUses AI   0.06383118 0.34565534
## CYears           0.01050201 0.02615455
## EdLevelBachelor's -0.26459520 0.04053397
## EdLevelMaster's  -0.18126969 0.15528839
## EdLevelProfessional/PhD -0.18644722 0.44203473
## EmploymentFreelance -0.34594785 0.17395122
## EmploymentPart-time -0.09194415 0.78552913
## EmploymentStudent -0.22968284 0.66878830
## Clog_comp        0.03895814 0.14856079
## RemoteWorkHybrid -0.03609734 0.30867880
## RemoteWorkRemote  0.07035527 0.42090775
```

Effects Plot (via MASS::polr)

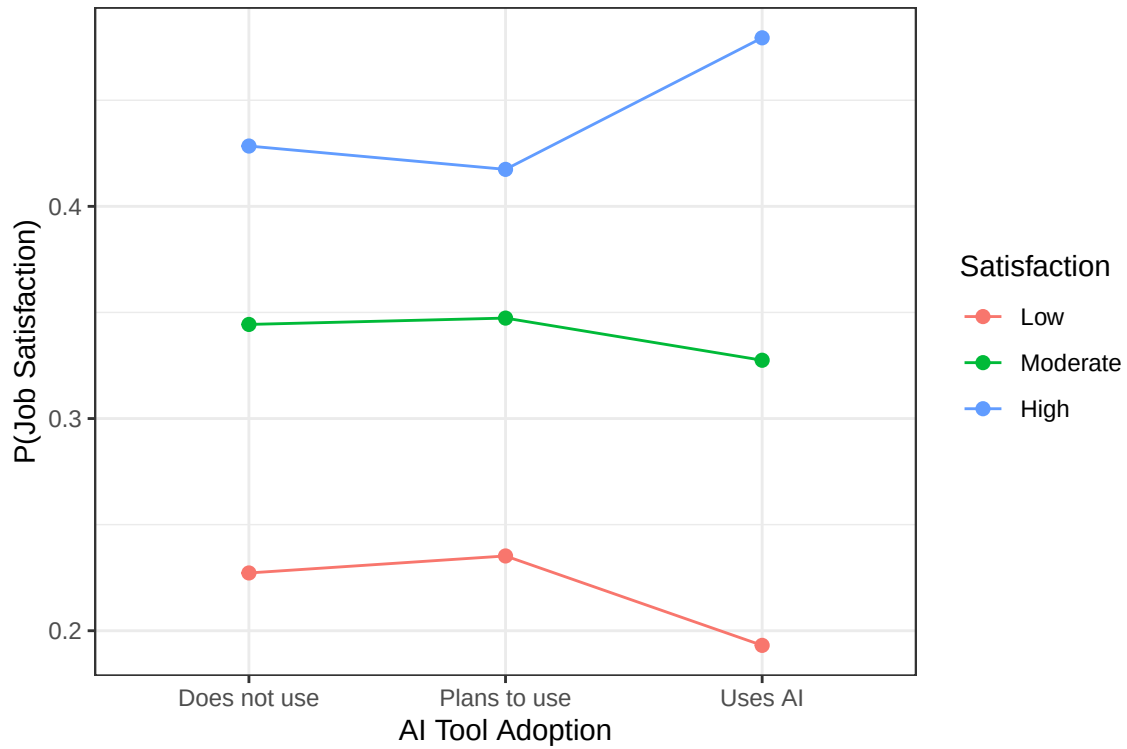
```
df_model <- df_model %>%
  mutate(across(where(is.factor), droplevels),
    JobSat_ord = ordered(JobSat_cat, levels = c("Low", "Moderate", "High")))

m_polr <- MASS::polr(JobSat_ord ~ AISelect + CYears + EdLevel +
  Employment + Clog_comp + RemoteWork,
  data = df_model, Hess = TRUE)

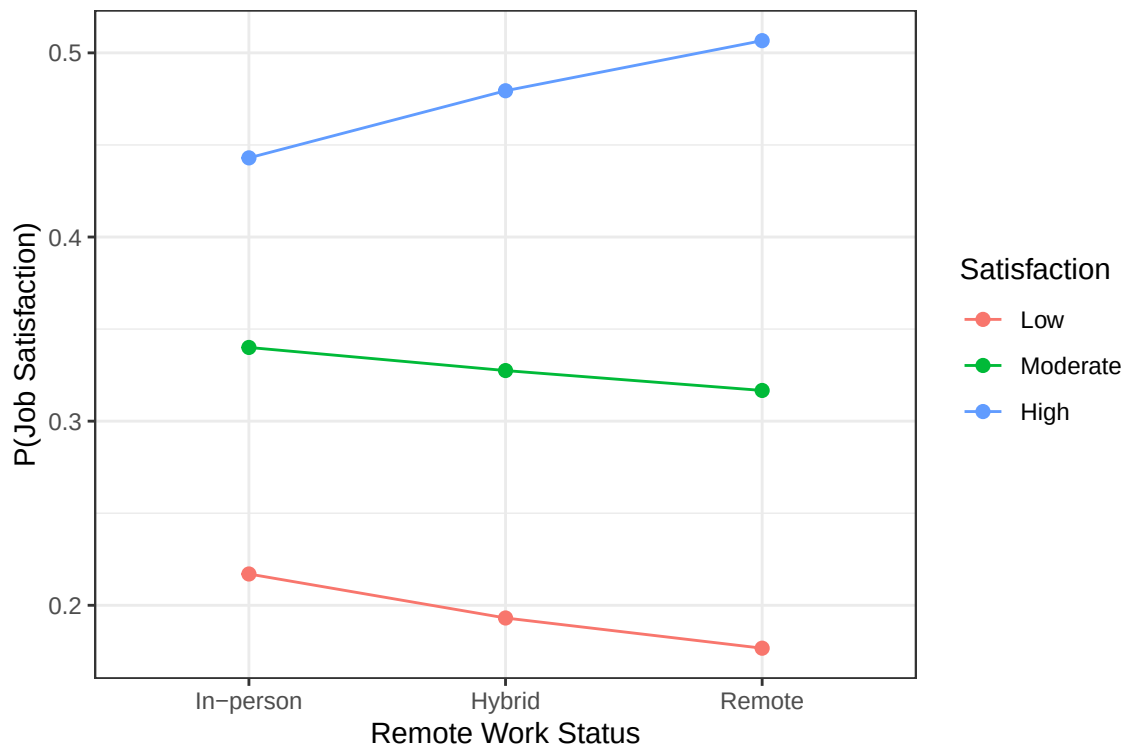
# Reference row held at means / most common category
ref_row <- data.frame(
  AISelect = factor("Uses AI", levels = levels(df_model$AISelect)),
  CYears = 0,
  EdLevel = factor("Bachelor's", levels = levels(df_model$EdLevel)),
  Employment = factor("Full-time", levels = levels(df_model$Employment)),
  Clog_comp = 0,
  RemoteWork = factor("Hybrid", levels = levels(df_model$RemoteWork))
)

eff_plot <- function(var, vals) {
  nd <- ref_row[rep(1, length(vals)), ]
  nd[[var]] <- vals
  rownames(nd) <- NULL
  cbind(nd[var],
    as.data.frame(predict(m_polr, newdata = nd, type = "probs"))) %>%
    pivot_longer(-all_of(var), names_to = "Satisfaction", values_to = "Prob") %>%
    mutate(Satisfaction = factor(Satisfaction, c("Low", "Moderate", "High")))
}
```

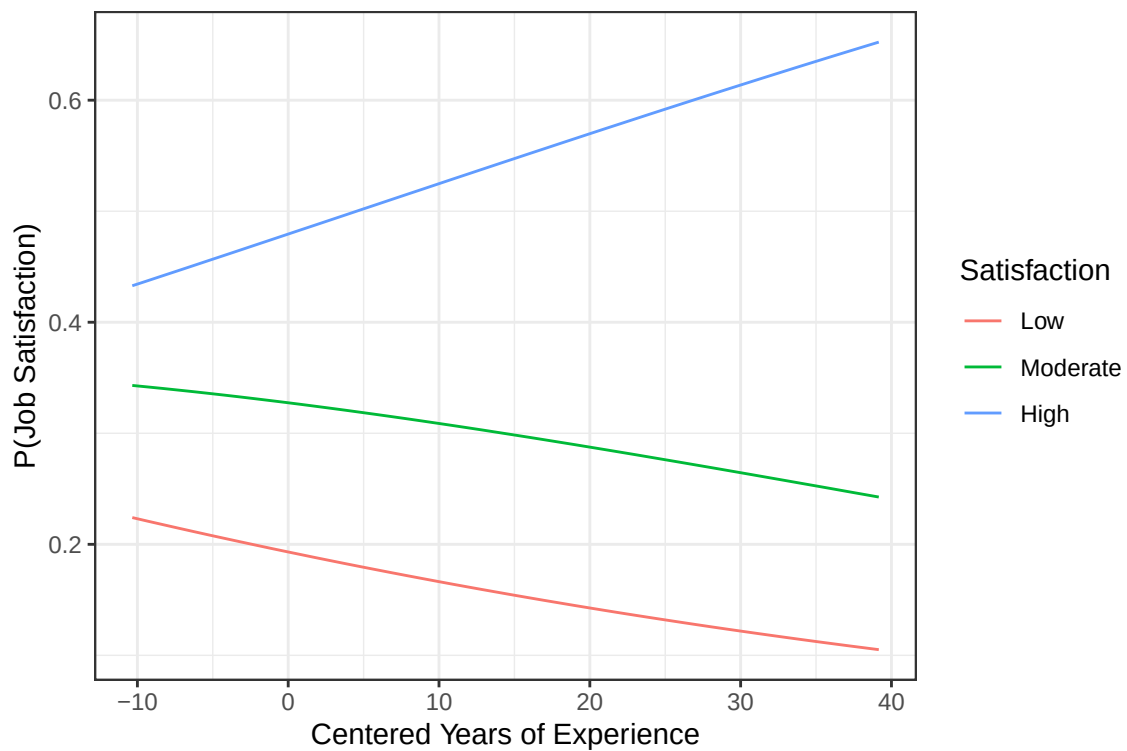
```
# AISelect
eff_plot("AISelect",
        factor(levels(df_model$AISelect), levels = levels(df_model$AISelect))) %>%
  ggplot(aes(x = AISelect, y = Prob, color = Satisfaction, group = Satisfaction)) +
  geom_line() + geom_point(size = 2) +
  labs(x = "AI Tool Adoption", y = "P(Job Satisfaction)") + theme_bw()
```



```
# RemoteWork
eff_plot("RemoteWork",
        factor(levels(df_model$RemoteWork), levels = levels(df_model$RemoteWork))) %>%
  ggplot(aes(x = RemoteWork, y = Prob, color = Satisfaction, group = Satisfaction)) +
  geom_line() + geom_point(size = 2) +
  labs(x = "Remote Work Status", y = "P(Job Satisfaction)") + theme_bw()
```



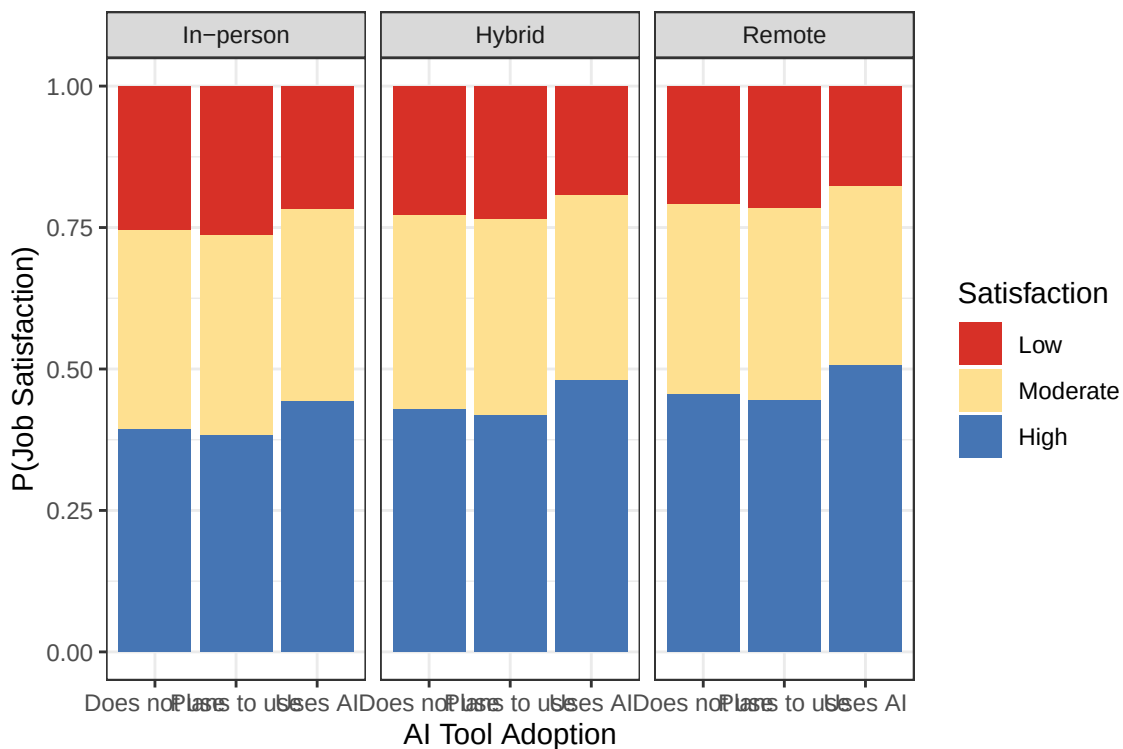
```
# CYears
eff_plot("CYears",
  seq(min(df_model$CYears), max(df_model$CYears), length.out = 50)) %>%
  ggplot(aes(x = CYears, y = Prob, color = Satisfaction)) +
  geom_line() +
  labs(x = "Centered Years of Experience", y = "P(Job Satisfaction)") + theme_bw()
```



```

expand.grid(
  AISelect = factor(levels(df_model$AISelect), levels = levels(df_model$AISelect)),
  RemoteWork = factor(levels(df_model$RemoteWork), levels = levels(df_model$RemoteWork)),
  CYears = 0,
  EdLevel = factor("Bachelor's", levels = levels(df_model$EdLevel)),
  Employment = factor("Full-time", levels = levels(df_model$Employment)),
  Clog_comp = 0
) %>%
  cbind(as.data.frame(predict(m_polr, newdata = ., type = "probs"))) %>%
  pivot_longer(c("Low", "Moderate", "High"),
               names_to = "Satisfaction", values_to = "Prob") %>%
  mutate(Satisfaction = factor(Satisfaction, c("Low", "Moderate", "High"))) %>%
  ggplot(aes(x = AISelect, y = Prob, fill = Satisfaction)) +
  geom_col(position = "stack") +
  facet_wrap(~ RemoteWork) +
  scale_fill_manual(
    values = c("Low" = "#d73027", "Moderate" = "#fee090", "High" = "#4575b4")
  ) +
  labs(x = "AI Tool Adoption", y = "P(Job Satisfaction)") + theme_bw()

```



Diagnostics

```

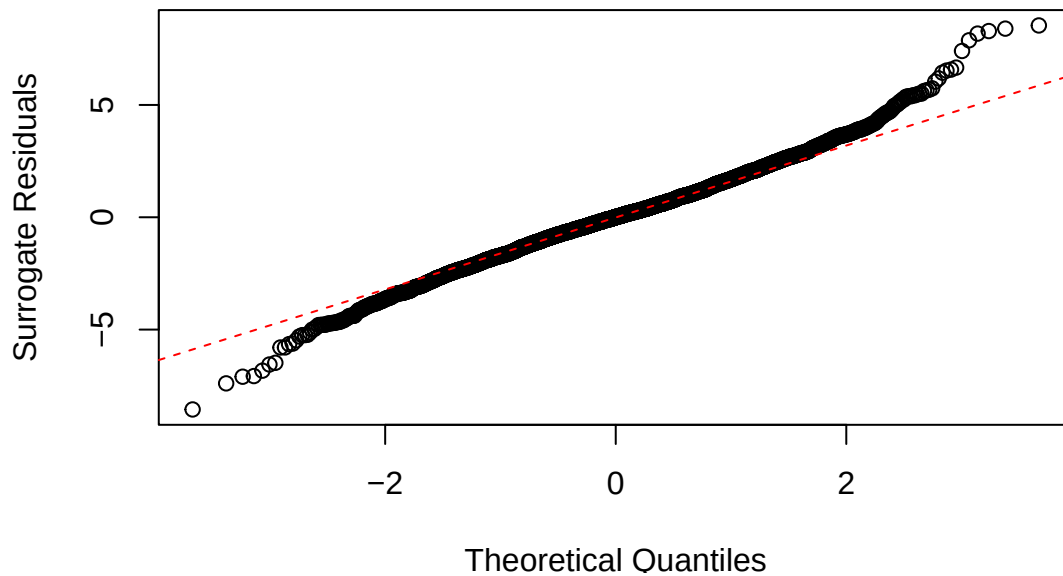
library(sure)
set.seed(123)
surr_resids <- sure::resids(m_polr, nsim = 50)

# Q-Q plot (replaces autoplot - sure's S3 method conflicts with ggplot2 3.x)
qqnorm(surr_resids,
       main = "Normal Q-Q Plot of Surrogate Residuals",
       ylab = "Surrogate Residuals")

```

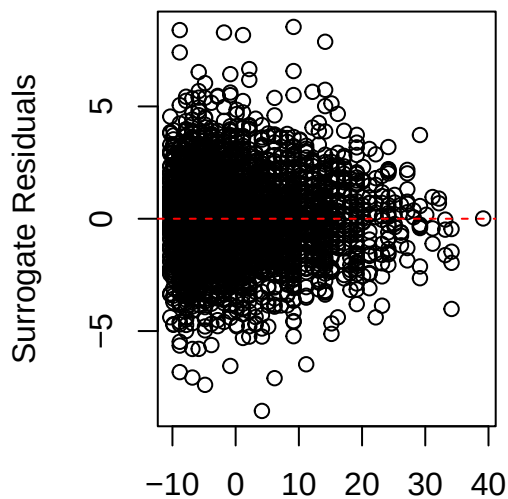
```
qqline(surr_resids, col = "red", lty = 2)
```

Normal Q-Q Plot of Surrogate Residuals

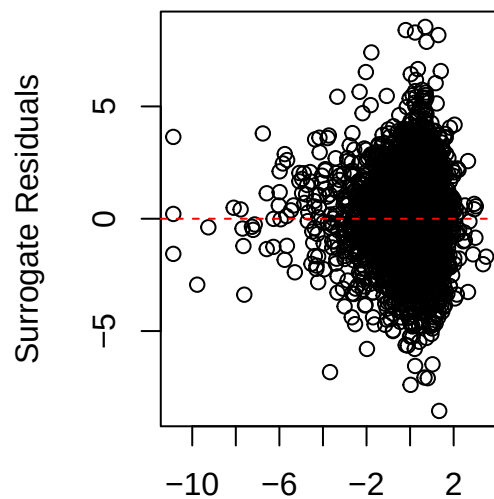


```
par(mfrow = c(1, 2))
plot(df_model$CYears, surr_resids,
     xlab = "Centered Years of Experience", ylab = "Surrogate Residuals")
abline(h = 0, col = "red", lty = 2)

plot(df_model$Clog_comp, surr_resids,
     xlab = "Centered Log Compensation", ylab = "Surrogate Residuals")
abline(h = 0, col = "red", lty = 2)
```



Centered Years of Experience



Centered Log Compensation

```
par(mfrow = c(1, 1))
```